Random or Not?

Name _____ Period _____ Date _____

What you do

1. Get one penny. Flip it ONE time. Keep the result a secret.

2. If that flip was HEADS, you will use GRID A for your real coin flips.

If that flip was TAILS, you will use GRID B for your real coin flips.

3. Flip the penny 100 times. Write H or T in each box of that grid, in order. Do not skip boxes.

4. In the OTHER grid, do not flip the coin. Use your brain. Write 100 letters (H or T) that LOOK random to you. Try to trick your teacher.

5. Write your secret answer in the box on the last page. Fold that part back so nobody can see it.

6. Give the paper to your teacher. Your teacher will try to say which grid is the real coin.

Do not write "real" or "fake" on the grids. Do not tell your partner. The two grids must look the same.

Words you need

STREAK: the same letter repeated in a row.

In H H T T T T H, the longest streak is 4. Four T's are together.

SWITCH: a place where the letter changes.

In H H T T T T H, there are 2 switches. H changes to T, then T changes to H.

RANDOM: nobody can predict the next one. Every flip is a fresh start.

Created with the help of Claude.ai (Anthropic).

Write one letter (H or T) in each box. The small gray numbers show where you are.

GRID A

1										
11										
21										
31										
41										
51										
61										
71										
81										
91										

GRID B

1										
11										
21										
31										
41										
51										
61										
71										
81										
91										

Part 1: Count your data

Count carefully. Use the words from page 1.

QUESTION	GRID A	GRID B
How many H?		
How many T?		
Longest streak of H?		
Longest streak of T?		
How many switches?		

Part 2: Count your runs

A RUN is a group of the same letter together. Look at this example:

H H T T T H T

This has 4 runs: H H (length 2), T T T (length 3), H (length 1), T (length 1).

So you would tally: one run of length 2, one run of length 3, two runs of length 1.

Now do the same for all 100 boxes in each grid. A run ends when the letter changes.

RUN LENGTH	HOW MANY IN GRID A	HOW MANY IN GRID B
1		
2		
3		
4		
5 or more		

CHECK: add up your runs. Both grids should have about 40 to 60 runs in total.

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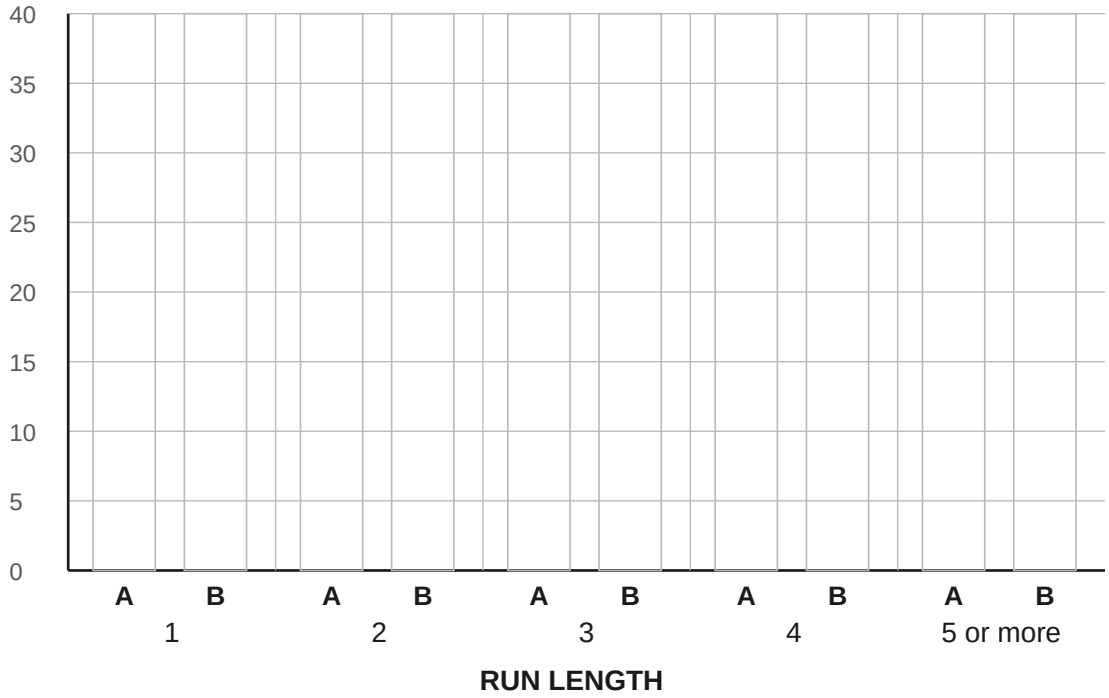
Part 3: Draw the bars

Shade one bar for each number in your table.
Bar A is for Grid A. Bar B is for Grid B. Shade up to the number you counted.

HOW MANY RUNS



Part 4: Class data

Your teacher will reveal which grid was the real coin. Then the class will add up everyone's runs on the board. Copy the class totals here.

RUN LENGTH	CLASS TOTAL: REAL	CLASS TOTAL: FAKE
1		
2		
3		
4		
5 or more		

Part 5: Think about it

1. Which grid do you think your teacher will pick as the real coin? Why? Answer this before your teacher tells you.

2. Was your teacher right? What do you think your teacher looked at?

3. When you made up the fake sequence, what rule were you using in your head? For example: "do not put too many H in a row." Write your rule.

4. In 100 real coin flips, the longest streak is usually 5, 6, or 7. Look at your table. Which grid is closer to that? Does this surprise you?

5. A real coin usually switches about 50 times in 100 flips. Which grid switched more? What does that tell you about how people fake randomness?

6. Look at the class totals. Which side had more SHORT runs (length 1 and 2)? Which side had more LONG runs (length 4 and 5 or more)?

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7. Why do people usually avoid writing long streaks when they try to fake randomness?

8. Do you think your teacher's guessing method will always work? Why or why not?

9. You flip a fair coin and get H H H H (four heads in a row). What is the chance that the next flip is H?

Circle one: less than 1/2 exactly 1/2 more than 1/2

10. Explain your answer to question 9. Does the coin remember the last four flips?

Part 6: Going further

11. A basketball player makes 6 shots in a row. Fans say she is "hot" tonight. After doing this activity, do you believe that? What else could explain 6 in a row?

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12. A student says: "I did not cheat on the test. My answers just happened to match my friend's answers." How could a teacher use ideas from today to check that claim?

13. Your music app has a "shuffle" button. It plays the same artist 3 times in a row and people say it is broken. Is the app broken? Explain.

14. If you flipped the coin 1,000 times instead of 100, what would you predict for the longest streak? Longer, shorter, or the same? Why?

15. Real randomness has more long streaks than people expect. Name one place in real life where this could fool somebody. Who gets hurt by being fooled?

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EXIT TICKET

Write one big idea you learned today about randomness.

_____

_____

FOLD ALONG THIS LINE SO YOUR TEACHER CANNOT SEE THE BOX BELOW

SECRET ANSWER (fill this in, then fold)

The REAL coin flips are in Grid: A B (circle one)

Teacher's guess: _____ Correct? YES NO

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